

No. of Functional Features Included in the Solution

Date	30-06-2026
Team ID	SWTID-2026-3845
Project Name	SMART LENDER
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.N o	Feature Name	Feature Description	Module/ Component	Status (Done / In Progress / Pending)	Marks Contribut ion
1	Data Preprocessing Pipeline	Cleans and prepares applicant data — encoding categorical fields, IQR-based outlier treatment, and SMOTE class balancing.	Data Preprocessing Module	Done	High
2	Model Training & Comparison	Trains and evaluates Decision Tree, KNN, Random Forest, Gradient Boosting, and XGBoost to select the best-performing classifier.	ML Training Module	Done	High
3	Model Serialization	Saves the selected best-performing model (XGBoost) and its scaler for reuse in real-time prediction.	Model Store (rdf.pkl, scale1.pkl)	Done	Medium
4	Applicant Submission Form	Web form allowing applicants to enter personal, financial, and loan-related details.	Flask Frontend (/submit)	Done	High
5	Real-Time Loan Prediction	Processes submitted data and returns an instant approval/rejection decision with a confidence score.	Flask Backend (/predict)	Done	High
6	Dashboard & Audit Log	Displays past predictions with applicant and outcome details for review by staff.	Flask Frontend (/)	Done	Medium
7	Database Integration	Stores applicant, credit, loan, model, and prediction data in a normalized MySQL schema.	MySQL Database	Done	Medium

8	Performance/Load Testing	Validates system behavior under concurrent user load using Locust.	Locust Testing Module	Done	Low
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	4
Additional / Nice-to-Have Features	4
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Submission Form, Dashboard & Audit Log
2	Backend / Logic	3	Data Preprocessing Pipeline, Model Training & Comparison, Model Serialization
3	Database / Storage	1	Database Integration (MySQL - Applicant, Loan, Credit, Model, Prediction tables)
4	API / Integration	2	Real-Time Loan Prediction (/predict endpoint), Performance/Load Testing (Locust)
5	Security / Authentication	0	Not yet implemented (planned for future release)